

Session 6 Quiz - Training Your Own AI Model

Track A | Grades 5-8

Q1 [Type: MCQ | Difficulty: Easy | QID:56962]

Complete the sentence: Kaggle is often compared to a _____ because it is a place where AI builders share and find various datasets.

- A. Hardware store
- B. Library
- C. Social media app
- D. Search engine

Correct Answer: B. Library

Explanation: Kaggle functions as a repository where datasets are stored and shared, much like books are kept in a library. AI builders can borrow or download datasets for their projects. Hardware store, social media app, and search engine do not capture the idea of a shared collection of resources.

Q2 [Type: MCQ | Difficulty: Easy | QID:56963]

Complete the sentence: The _____ set is like the 'real exam' for the AI because it contains photos it has never seen during its study time.

- A. Kaggle
- B. Training
- C. Practice
- D. Test

Correct Answer: D. Test

Explanation: The test set is used to evaluate the model's performance on new, unseen data to see if it can generalize what it learned. Training data is for studying, practice is not a standard term, and Kaggle is a platform.

Q3 [Type: MCQ | Difficulty: Easy-Medium | QID:56964]

Complete the sentence: In AI training, a single _____ represents one full round of the AI going through every single photo in the training set.

- A. Experiment
- B. Class
- C. Dataset
- D. Epoch

Correct Answer: D. Epoch

Explanation: An epoch is the technical term for one complete pass of the training algorithm through the entire dataset. In Teachable Machine's advanced settings, you can adjust the number of epochs. Experiment, class, and dataset are not correct terms for this concept.

Q4 [Type: MCQ | Difficulty: Easy-Medium | QID:56965]

Complete the sentence: If an AI simply memorizes the training photos instead of learning actual patterns, it is experiencing _____.

- A. Overfitting
- B. Under-training
- C. Generalization
- D. Data variety

Correct Answer: A. Overfitting

Explanation: Overfitting occurs when a model learns the specific details and noise in the training data to the point that it hurts performance on new data. The AI memorizes rather than generalizes. Under-training is not enough learning, generalization is the goal, and data variety helps prevent overfitting.

Q5 [Type: MCQ | Difficulty: Easy | QID:56966]

Complete the sentence: The principle of '_____ in, garbage out' means that if you train an AI with bad data, it will produce bad results.

- A. Data
- B. Photos
- C. Garbage
- D. Coding

Correct Answer: C. Garbage

Explanation: The full phrase is "Garbage In, Garbage Out." It emphasizes that the quality of the output is directly limited by the quality of the input data. Data, photos, and coding do not complete the rhyming phrase correctly.

Q6 [Type: True/False | Difficulty: Easy | QID:56967]

True or False: Using thousands of photos of only golden retrievers is the best way to train an AI to recognize all types of dogs.

- A. True
- B. False

Correct Answer: B. False

Explanation: An AI needs a variety of breeds, colors, and sizes to learn a general concept of what a dog looks like. Only golden retrievers would cause the AI to miss other breeds entirely. Diversity in training data is essential.

Q7 [Type: True/False | Difficulty: Easy | QID:56968]

True or False: AI builders use Kaggle because it is much faster than taking thousands of photos themselves.

A. False

B. True

Correct Answer: B. True

Explanation: Kaggle provides ready-made datasets, which saves engineers the immense time it would take to collect, photograph, and label thousands of examples manually. It accelerates the AI development process significantly.

Q8 [Type: True/False | Difficulty: Easy | QID:56969]

True or False: In Teachable Machine, 'Training' is the stage where the AI looks for patterns in the photos you uploaded.

A. False

B. True

Correct Answer: B. True

Explanation: During the training phase, the algorithm analyzes the data to establish the mathematical relationships that define each class. This happens after you upload photos but before you test the model.

Q9 [Type: True/False | Difficulty: Easy-Medium | QID:56970]

True or False: If you train an AI for more epochs, it will always get more accurate on the test photos.

A. False

B. True

Correct Answer: A. False

Explanation: While more training can help, there is a limit where the AI might start memorizing instead of learning (overfitting), which can hurt performance on test data. More epochs do not guarantee better results.

Q10 [Type: True/False | Difficulty: Easy-Medium | QID:56971]

True or False: To make a fair comparison between experiments, you should change your test photos every time you train a new model.

A. False

B. True

Correct Answer: A. False

Explanation: Keeping test photos the same allows you to see if changes in the training data actually made the AI smarter. Changing the test set would introduce a new variable, making comparisons unfair.

Q11 [Type: MCQ | Difficulty: Easy-Medium | QID:56972]

What is the likely result of training an AI with only 1 photo of a cat and 1 photo of a dog?

- A. The AI will likely guess or memorize that specific photo's background.
- B. The AI will be able to recognize any cat or dog accurately.
- C. The AI will perfectly generalize to all breeds.
- D. The AI will automatically download more photos from the internet.

Correct Answer: A. The AI will likely guess or memorize that specific photo's background.

Explanation: With very little data, the AI has no way to find general patterns and will rely on irrelevant details like background color or lighting. It cannot recognize new cats or dogs accurately.

Q12 [Type: MCQ | Difficulty: Easy-Medium | QID:6973]

Which of these is a reason why AI builders need variety in their training backgrounds?

- A. Because Teachable Machine requires at least five different colors.
- B. So the AI doesn't think a 'cat' is just any animal sitting on a green rug.
- C. So the AI can learn to identify the furniture instead of the animal.
- D. To make the photos look more professional and artistic.

Correct Answer: B. So the AI doesn't think a 'cat' is just any animal sitting on a green rug.

Explanation: If all training cats are on green rugs, the AI might mistakenly learn that the rug is the defining feature of a cat. Variety in backgrounds helps the AI focus on the actual object (the cat) instead of the setting.

Q13 [Type: MCQ | Difficulty: Easy-Medium | QID:56974]

In the 'Cats vs Dogs' dataset structure mentioned in the text, where are the 'unseen' photos kept?

- A. The Kaggle homepage
- B. The advanced settings menu
- C. The test_set folder
- D. The training_set folder

Correct Answer: C. The test_set folder

Explanation: The test set is specifically reserved for evaluation and contains images the AI does not see during training. It acts as the exam. The training_set folder holds the practice photos.

Q14 [Type: MCQ | Difficulty: Easy-Medium | QID:56975]

If you increase the number of training photos from 10 to 100 for each category, what is the most likely outcome?

- A. The AI will find it harder to tell the difference between animals.
- B. The AI will become better at recognizing new photos.
- C. The AI will get slower and less accurate.
- D. The AI will stop using patterns and start guessing.

Correct Answer: B. The AI will become better at recognizing new photos.

Explanation: More data provides more variety, which helps the AI learn the real features of cats and dogs. This usually leads to better generalization and higher accuracy on new, unseen photos.

Q15 [Type: MCQ | Difficulty: Easy-Medium | QID:56976]

What is the purpose of the 'Advanced' settings in Teachable Machine used in Experiment 4?

- A. To adjust how many times the AI studies the data (epochs).
- B. To delete photos that the AI got wrong.
- C. To upload the photos to Kaggle.
- D. To change the colors of the cats and dogs.

Correct Answer: A. To adjust how many times the AI studies the data (epochs).

Explanation: The advanced settings allow users to tweak training parameters, such as the number of epochs (how many times the AI reviews the entire dataset). This lets you experiment with different training durations.

Q16 [Type: MCQ | Difficulty: Medium | QID:56977]

Why is it important to include different angles (front, side, looking away) in your training data?

- A. To teach the AI how to rotate images automatically.
- B. So the AI knows a cat is still a cat, even if it's not looking at the camera.
- C. To make the AI training process faster.
- D. Because the test photos will always be taken from the side.

Correct Answer: B. So the AI knows a cat is still a cat, even if it's not looking at the camera.

Explanation: Variety in angles helps the AI learn the 3D features of the animal rather than just a 2D silhouette. This ensures the AI recognizes a cat from any viewpoint, not just the ones it memorized.

Q17 [Type: MCQ | Difficulty: Easy-Medium | QID:56978]

According to the session, how does a 'Real AI' used by professional engineers compare to your Teachable Machine experiment?

- A. Real AI only works with numbers, never with photos.
- B. Real AI doesn't need a test set because it's always right.
- C. Real AI uses fewer photos to save money.
- D. Real AI takes days or weeks to train instead of seconds.

Correct Answer: D. Real AI takes days or weeks to train instead of seconds.

Explanation: Because of the massive amount of data and complex mathematical computations, professional AI training is a very long process (days or weeks), unlike the quick training in Teachable Machine.

Q18 [Type: MCQ | Difficulty: Medium | QID:56979]

If your AI has a 99% confidence score on a training photo but only 50% on a test photo, what does this suggest?

- A. The AI has generalized perfectly.
- B. You need to decrease the number of training photos.
- C. The test photos are broken or corrupted.
- D. The AI is overfitted and memorized the training data.

Correct Answer: D. The AI is overfitted and memorized the training data.

Explanation: The model performs well on what it saw before (training data) but fails on new examples (test data). This is a classic sign of overfitting, meaning it memorized rather than learned general patterns.

Q19 [Type: MCQ | Difficulty: Easy-Medium | QID:56980]

What is the primary benefit of the Kaggle community for AI fans?

- A. It provides a place to share knowledge and compete.
- B. It automatically builds your AI models for you for free.
- C. It is a social network for pets.
- D. It is the only place to buy AI hardware.

Correct Answer: A. It provides a place to share knowledge and compete.

Explanation: Kaggle fosters a community where people can learn from each other, share datasets and code, and test their skills in competitions. It does not automatically build models, is not for pets, and does not sell hardware.

Q20 [Type: MCQ | Difficulty: Medium | QID:56981]

Which experiment in the session proved that 'More Data = Smarter AI'?

- A. Experiment 1 (1 photo)
- B. Experiment 4 (Same data, longer training)
- C. The Homework assignment
- D. Experiment 3 (~100 photos)

Correct Answer: D. Experiment 3 (~100 photos)

Explanation: Experiment 3 demonstrated a big jump in accuracy when the amount of data was significantly increased compared to earlier experiments with very few photos. More data provided more variety and led to a smarter AI.